

# Large Language Models for Socio-Political Event Interpolation and Extrapolation

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**Abstract.** Socio-political events are commonly represented as temporal knowledge graph (TKG) quadruples in the format of (subject, relation, object, timestamp), and can be extended to quintuples by appending auxiliary textual summaries. Existing event prediction methods predominantly rely on graph neural networks (GNNs) and recurrent neural networks (RNNs) to capture structural and sequential patterns, yet they under-utilize the rich semantic information embedded in event descriptions. In this paper, we propose LEAP, a systematic framework that leverages large language models (LLMs) as event predictors for both interpolation and extrapolation tasks. For interpolated object prediction (OP), we introduce two complementary formulations: (i) a ranking approach that integrates fine-tuned encoder representations of textual summaries with structural embeddings through a convolutional decoder, and (ii) a generative approach that frames OP as instruction-tuned question answering with carefully designed prompt templates. For extrapolated multi-event forecasting (MEF), we propose a quintuple-level prompt engineering strategy that circumvents the limited context windows of open-source language modeling encoders, combined with an encoder-agnostic and aggregation-agnostic architecture supporting multiple embedding aggregation mechanisms. Extensive experiments on three geographically distinct ICEWS datasets demonstrate that LEAP achieves state-of-the-art performance on both tasks. LEAP’s code, sample data, and appendix are available at <https://github.com/yuening-lab/LEAP/>.

**Keywords:** Temporal knowledge graphs · Socio-political event prediction · Spatial-temporal modeling · Large language models.

## 1 Introduction

The ability to accurately predict socio-political events, such as diplomatic interactions, policy decisions, and interstate conflicts, is critically important for proactive decision-making, risk assessment, and resource allocation [44, 41]. These events are commonly encoded as temporal knowledge graph (TKG) quadruples  $(s, r, o, t)$  denoting a subject  $s$ , relation  $r$ , object  $o$ , and timestamp  $t$ , and can be augmented with textual summaries  $x$  to form quintuples  $(s, r, o, t, x)$  [6, 23]. Event prediction on TKGs encompasses two canonical settings: interpolation,

where one infers the missing object entity given observed query subject, relation, and timestamp, and extrapolation, where one forecasts multiple possibly occurred relations in the future without anchored queries [40, 7]. Contemporary state-of-the-art frameworks address these tasks through intricate combinations of graph neural networks (GNNs) [31], which capture spatially relational structure among entities and recurrent neural networks (RNNs) [30], which model temporal dynamics between history and future [18, 11, 16]. While effective, these architectures exhibit two notable shortcomings. First, they largely ignore the rich semantic context available in textual event descriptions, relying instead on discrete entity and relation identifiers or, at best, pre-trained topic-model clusters [23]. Second, their composite design, typically stacking specialized encoders for entities, relations, and temporal patterns, introduces considerable engineering complexity and limits transferability across tasks.

Large language models (LLMs) offer a compelling opportunity to address both shortcomings, given their demonstrated proficiency in semantic understanding, contextual reasoning, and few-shot generalization [29, 34]. However, applying LLMs to TKG-based event prediction is non-trivial. We identify two concrete technical challenges that motivate the design of our framework.

*Challenge 1: Scale-sensitive cost of LLM-based event prediction.* Socio-political event datasets such as ICEWS [6] contain millions of events, each requiring contextual reasoning over structured quadruples and textual summaries. Prompting proprietary models such as GPT-4o [25] and GPT-5.2 [26] for every query incurs prohibitive financial overhead at this scale [42, 27], yet conventional GNN/RNN pipelines forgo the semantic understanding that LLMs provide.

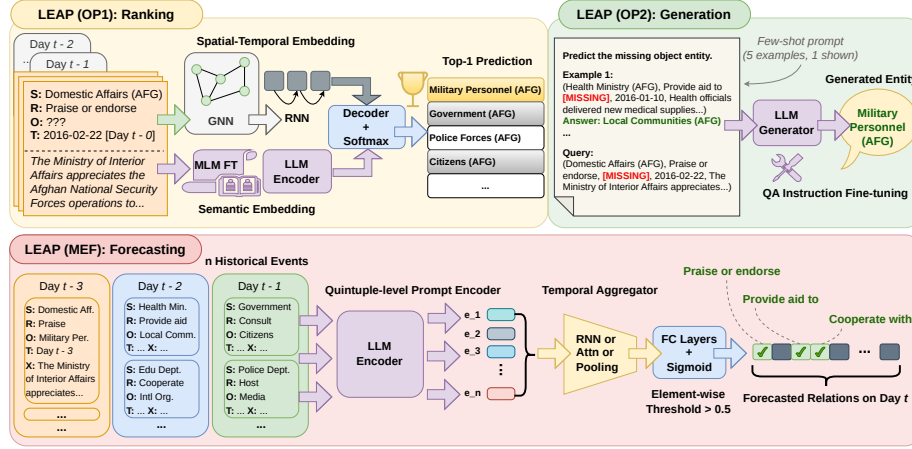
*Challenge 2: Bridging textual semantics and structural knowledge.* TKG prediction benefits from both the relational inductive bias of graph-based representations and the contextual richness of natural language. However, existing methods treat these modalities largely in isolation [7], and naively concatenating daily event histories either triggers truncation or degrades model performance [17].

To address these challenges, we propose LEAP (Large language models for Extrapolated and interpolAted socio-political event Predictions), a systematic investigation framework encompassing three complementary components.

1. LEAP<sub>OP1</sub> formulates object prediction as a ranking task. We fine-tune an encoder-only pre-trained language model, RoBERTa<sub>BASE</sub> [22], on domain-specific event text with masked language modeling objective, and fuse the encoded textual summary embeddings with structural entity and relation embeddings, obtained from R-GCN [32] and GRU [8], through a convolutional decoder ConvTransE [33] to rank all candidate objects.
2. LEAP<sub>OP2</sub> formulates object prediction as a generative task. We design multiple prompt templates that cast object prediction as instruction-tuned question answering, and fine-tune an encoder-decoder pre-trained language model (FLAN-T5<sub>BASE</sub> [9]) to directly generate the missing object entity.

3.  $\text{LEAP}_{\text{MEF}}$  formulates multi-event forecasting as multi-label binary classification task. We implement quintuple-level prompt engineering, where each quintuple is independently encoded by a pre-trained encoder-only language model, and the resulting embeddings are aggregated through a configurable aggregation module, supporting self-attention [39], Long Short-Term Memory (LSTM) [14], mean pooling, and max pooling, followed by a Sigmoid-activated fully connected layer to predict possible future relation occurrences. This encoder-agnostic and aggregation-agnostic design allows practitioners to substitute different encoder-only language models and aggregation strategies without modifying the overall architecture.

Extensive experiments on three geographically, socially, and politically distinct country subsets of the ICEWS dataset [6] (Afghanistan, India, Russia) demonstrate LEAP achieves state-of-the-art results on both OP and MEF across diverse evaluation metrics with manageable computational cost. Notably, our fine-tuned compact models consistently outperform substantially larger frontier LLMs, including DeepSeek-V3.2 [21], GPT-4o [25], and GPT-5.2 [26], which are used in a prompting-only regime, indicating that task-specific adaptation is more effective than model scaling for domain-specific structured prediction.



**Fig. 1.** LEAP framework overview.  $\text{LEAP}_{\text{OP1}}$  combines spatial-temporal embeddings from R-GCN and GRU with semantic embeddings from a masked language modeling (MLM) fine-tuned (FT) RoBERTa encoder, decoded by ConvTransE for ranking-based object prediction.  $\text{LEAP}_{\text{OP2}}$  formulates object prediction as question answering (QA) by instruction fine-tuning FLAN-T5 with few-shot prompt templates to directly generate the missing object entity.  $\text{LEAP}_{\text{MEF}}$  encodes each historical quintuple independently using a pre-trained language modeling encoder, DeBERTaV3, integrates the resulting embeddings through an attention-based (Attn), RNN-based, or mean/max pooling-based temporal aggregator, and applies element-wise Sigmoid-activated fully connected (FC) layers to forecast possible future relation occurrences.

## 2 Related Work

Our work builds on two lines of research: temporal knowledge graph reasoning (TKGR) and LLMs’ applications to structured prediction tasks. Existing TKGR methods predominantly rely on GNN and RNN architectures that treat textual event descriptions as auxiliary signals processed through shallow encoders or topic models [18, 11, 16, 23], while recent LLM-based approaches have explored event prediction through few-shot reasoning [35] and generative forecasting [19] but have not systematically investigated LLMs as direct or auxiliary predictors for both interpolated and extrapolated TKGR tasks. Due to page limit, a detailed discussion of related work is provided in supplementary material Appendix A.

## 3 Methodology

We first formalize two tasks, object prediction (OP) and multi-event forecasting (MEF). We then describe three complementary LEAP components in detail.

### 3.1 Problem Formulation

Let  $\mathcal{V}$  denote the set of entities,  $\mathcal{R}$  the set of relation types, and  $\mathcal{T}$  the set of timestamps. A temporal knowledge graph is a collection of quintuples  $\{(s, r, o, t, x)\}$ , where  $s, o \in \mathcal{V}$ ,  $r \in \mathcal{R}$ ,  $t \in \mathcal{T}$ , and  $x$  is a textual summary.

*Object Prediction (OP).* Given an incomplete query quintuple  $(s_q, r_q, ?, t_q, x_q)$  where the object entity is missing and a list of historical events  $\mathcal{G}_{\leq t_q}$  as background knowledge, the goal is to identify the correct object  $o_q \in \mathcal{V}$ .

*Multi-Event Forecasting (MEF).* Given a list of historical events  $\mathcal{G}_{< t_n}$  up to but excluding a future timestamp  $t_n$  as background knowledge, the goal is to predict which relations  $r_{t_n}^i \in \mathcal{R}$  ( $i = 1, \dots, |\mathcal{R}|$ ) could possibly occur at  $t_n$ .

### 3.2 LEAP<sub>OP1</sub>: Object Prediction via Ranking

As shown in Figure 1, LEAP<sub>OP1</sub> integrates spatial-temporal and semantic embeddings through a three-stage pipeline.

*Stage 1: Spatial-Temporal Embedding Learning.* For each historical snapshot in  $\mathcal{G}_{\leq t_q}$ , we employ R-GCN [32] to spatially update entity embeddings  $\mathbf{E}_{\mathcal{V}}$  following graph structure at each historical timestamp and GRU [8] to maintain temporally evolving relation embeddings  $\mathbf{E}_{\mathcal{R}}$  over a historical window of length  $l_1$ .

*Stage 2: Semantic Embedding Learning.* We collect textual summaries from all training quintuples and fine-tune RoBERTa<sub>BASE</sub> [22] using the standard masked language modeling objective. This domain-adaptive pre-training enriches the encoder with event-specific linguistic patterns. For each summary  $x_i$ , we obtain a sentence-level embedding  $\mathbf{e}_i^{\text{text}} = \text{MeanPool}(\text{RoBERTa}(x_i))$ .

*Stage 3: Object Probability Estimation.* We employ ConvTransE [33] as the decoding module, which integrates spatial-temporal embeddings  $\mathbf{E}_{\mathcal{V},\mathcal{R}}$  with semantic embeddings  $\mathbf{E}_{\text{text}}$  through linear projection, non-linear activation, and convolutional scoring, followed by the Softmax normalization over all candidate entities to obtain probabilities for ranking evaluation:

$$\mathbf{P}_q = \text{Softmax}[\text{ConvTransE}(\mathbf{E}_{\mathcal{V},\mathcal{R}}, \mathbf{E}_{\text{text}})] \Big|_q \quad (1)$$

where  $\mathbf{P}_q \in \mathbb{R}^{b \times |\mathcal{V}|}$ ,  $b$  is the query batch size, and  $\sum_{j=1}^{|\mathcal{V}|} p_{q_i}^j = 1$  for each query event  $q_i$ . Candidate entity with the highest probability is selected as the predicted object. The training of LEAP<sub>OP1</sub> optimizes R-GCN, GRU, and ConvTransE jointly through cross-entropy loss; the RoBERTa<sub>BASE</sub> encoder is fine-tuned separately in Stage 2 to ensure stable convergence during domain adaptation.

### 3.3 LEAP<sub>OP2</sub>: Object Prediction via Generation

LEAP<sub>OP2</sub> re-casts OP as an instruction-tuned question-answering (QA) task. We design three prompt templates to systematically investigate the contributions of in-context examples and textual summaries, and we evaluate predictions using ROUGE scores [20] that measure unigram (R-1), bigram (R-2), and longest common subsequence (R-L) overlap between generated and ground-truth objects. We adopt ROUGE rather than exact match because socio-political entity names often admit surface-level variation such as abbreviations, and ROUGE captures partial token overlap that strict string matching would penalize.

*Few-Shot Prompt.* The few-shot input prompt  $Q_{\text{in}}$  starts with clear instructions, followed by five historical quintuples formatted as in-context learning examples. Each example presents a quintuple in the format (subject, relation,  $\langle \text{MISSING OBJECT ENTITY} \rangle$ , timestamp, text summary) with its ground-truth object revealed. A query event quintuple with the object masked is then appended, along with the instruction to predict the missing object entity. The complete few-shot prompt template is illustrated in supplementary material Appendix B.

*Zero-Shot Prompt.* All five in-context learning examples are removed from the few-shot prompt, retaining only the incomplete query and task instructions.

*No-Text Prompt.* All textual summaries are removed from both in-context learning examples and the query event in the few-shot prompt, isolating the contribution of structural quadruple information.

We employ FLAN-T5<sub>BASE</sub> [9], which is an encoder-decoder pre-trained language model with approximately 250 million (M) parameters. We fine-tune it using standard cross-entropy objective in a sequence-to-sequence format, where the input sequence is the complete prompt, and the expected output sequence is the ground-truth object entity.

*Rationale for Model Selection.* We deliberately select compact and open-source models (RoBERTa<sub>BASE</sub> with  $\sim 125\text{M}$  parameters; FLAN-T5<sub>BASE</sub> with  $\sim 250\text{M}$  parameters) for two reasons. First, reproducibility and accessibility: these models can be fine-tuned on a single commodity GPU, enabling independent verification without proprietary API access. Second, empirical justification: as experimental results demonstrate in subsection 4.3, task-specific instruction fine-tuning of these open-source compact models consistently outperforms substantially larger commercialized models in a prompting-only regime.

*Ranking vs. Generation Deployment.* The two OP formulations target complementary deployment scenarios along several practical dimensions. *Output modality:* LEAP<sub>OP1</sub> produces a calibrated probability distribution over all entities, making it preferable in retrieval or decision-support settings where multiple plausible objects should be surfaced and ranked; LEAP<sub>OP2</sub> outputs a single human-readable entity name, better suited to report generation or analyst briefing workflows. *Output coverage:* LEAP<sub>OP1</sub> is restricted to entities observed during training (closed-world assumption), whereas LEAP<sub>OP2</sub> can in principle generate novel entity names unseen in the training set (open-world capability). *Failure modes:* LEAP<sub>OP1</sub> may assign high probability to structurally plausible but semantically incorrect entities when the graph neighborhood is ambiguous, while LEAP<sub>OP2</sub> may generate fluent but hallucinated entity names that do not correspond to any real actor, particularly for rare or unseen query patterns. Researchers should select between the two formulations based on whether their application prioritizes ranked candidate lists or open-world generalization.

### 3.4 LEAP<sub>MEF</sub>: Multi-Event Forecasting

Based on Figure 1, LEAP<sub>MEF</sub> formulates MEF as a multi-label binary classification problem over the relation set  $\mathcal{R}$ , predicting which relations could possibly occur at a future timestamp  $t_n$ . A key design principle is modularity: LEAP<sub>MEF</sub> is both encoder-agnostic and aggregation-agnostic, allowing framework users to substitute different encoder-only LLMs and temporal aggregation strategies without significantly modifying the overall pipeline.

*Quintuple-Level Prompt Engineering.* Given limited context windows for many open-source encoder-only language models [2], we process each quintuple independently. For each quintuple, we construct a simple structured prompt  $u_i$  that concatenates the five elements (subject, relation, object, timestamp, text summary) as labeled fields, and obtain a fixed-dimensional embedding through a pre-trained encoder-only language model with mean pooling:

$$\mathbf{e}_i = \text{MeanPool}(\text{Encoder}(u_i)) \quad (2)$$

where  $\mathbf{e}_i \in \mathbb{R}^{d_h}$  and  $d_h$  is the hidden dimension of the chosen encoder. This quintuple-level encoding strategy decouples the number of historical events from the encoder’s maximum input context limit, enabling the framework to incorporate arbitrarily large event histories without truncation. Notably, LEAP<sub>MEF</sub>

does not explicitly model entity-relation graph connectivity. We hypothesize that this is viable for two reasons: first, the textual summaries accompanying each quintuple implicitly encode relational context (e.g., actor roles, event descriptions) that a pre-trained language model can leverage without explicit graph structure; second, MEF predicts relation-type occurrences rather than specific entity-level links, a coarser-grained task for which sequential patterns in recent event histories provide sufficient signal, as corroborated by the sensitivity analysis in subsection 4.5. In our experiments, we evaluate two encoder choices: RoBERTa<sub>LARGE</sub> [22] and DeBERTaV3<sub>LARGE</sub> [13].

*Day-Level Aggregation.* Quintuple embeddings are grouped by timestamps to form daily embedding matrices  $\mathbf{E}_j \in \mathbb{R}^{n_j \times d_h}$ , where  $n_j$  is the number of events on day  $j$ . Given a historical window of length  $l_3$ , we concatenate daily embeddings and apply one of the following aggregation mechanisms:

1. Self-Attention (SA) [39]: A self-attention layer followed by mean aggregation over the historical time dimension.
2. LSTM [14]: An LSTM module that processes the sequence of quintuple embeddings. The final hidden state serves as the aggregated representation.
3. Mean Pooling: Element-wise averaging across all quintuple embeddings.
4. Max Pooling: Element-wise maximum across all quintuple embeddings.

For the self-attention variant, the aggregation is formulated as:

$$\tilde{\mathbf{E}}_{t_n} = \text{MeanPool}[\text{SelfAttn}(\mathbf{E}_{t_n-l_3 \leq t < t_n})] \Big|_{t_n} \quad (3)$$

where  $\tilde{\mathbf{E}}_{t_n} \in \mathbb{R}^{b \times d_h}$  and  $b$  is the batch size. Similarly, for the LSTM variant, the aggregation is formulated as:

$$\tilde{\mathbf{E}}_{t_n} = \text{LSTM}(\mathbf{E}_{t_n-l_3 \leq t < t_n}) \Big|_{t_n} \quad (4)$$

*Relation Occurrence Prediction.* A fully connected layer with learnable parameters  $\mathbf{W}_z \in \mathbb{R}^{d_h \times |\mathcal{R}|}$  and bias  $\mathbf{b}_z \in \mathbb{R}^{|\mathcal{R}|}$  maps the aggregated embeddings to relation-level scores, followed by element-wise Sigmoid activation:

$$\mathbf{P}_{t_n} = \sigma(\mathbf{W}_z^\top \tilde{\mathbf{E}}_{t_n} + \mathbf{b}_z) \quad (5)$$

where  $\mathbf{P}_{t_n} \in \mathbb{R}^{b \times |\mathcal{R}|}$  represents the probability of event occurrences. The training of LEAP<sub>MEF</sub> optimizes the aggregation module and the fully connected layer jointly through the binary cross-entropy objective.

## 4 Experiments

### 4.1 Datasets and Implementations

We conduct experiments on three country-specific subsets of the Integrated Crisis Early Warning System (ICEWS) dataset [6]: Afghanistan (AFG), India (IND),

**Table 1.** ICEWS [6] subset statistics.  $o \in x$  (%) indicates the percentage of event quintuples whose textual summary contains exactly the object entity name.

| Dataset     | $ \mathcal{V} $ | $ \mathcal{R} $ | $ \mathcal{Q} _{\text{train}}$ | $ \mathcal{Q} _{\text{valid}}$ | $ \mathcal{Q} _{\text{test}}$ | $o \in x$ (%) |
|-------------|-----------------|-----------------|--------------------------------|--------------------------------|-------------------------------|---------------|
| Afghanistan | 3,756           | 218             | 212,540                        | 32,734                         | 34,585                        | 20.7          |
| India       | 6,298           | 234             | 318,471                        | 75,439                         | 85,739                        | 19.5          |
| Russia      | 7,798           | 237             | 275,477                        | 46,516                         | 51,371                        | 45.3          |

and Russia (RUS). Table 1 summarizes key statistics, where more details are described in supplementary material Appendix C. The three subsets are selected to provide geopolitical diversity, and to ensure comparability with prior work [11]. To illustrate the quintuple format, we present an example from the AFG test set: (Taliban, Use conventional military force, Military (Afghanistan), 2016-02-22, “Two soldiers of local police were killed as a result of a Taleban attack in Shah Joy District of Zabol Province [southern Afghanistan] and three soldiers of the Afghan National Army [ANA] killed in Dasht-e Archi District of Konduz Province [northern Afghanistan].”). The textual summary does not contain the object entity, but does provide rich contextual detail that goes beyond what the structured quadruple elements encode, motivating our use of language models to capture this semantic signal. An important empirical question is whether the textual summary  $x$  directly reveals the object entity  $o$ , which would reduce object prediction to surface-level entity extraction. We measure the proportion of quintuples in which the object entity name appears in the textual summary and find rates of 20.7% for AFG and 19.5% for IND, confirming OP requires genuine relational reasoning rather than simple extraction for the majority of events.

All experiments involving training and inference are conducted on a Linux Ubuntu server equipped with  $4 \times 48$  GB NVIDIA L40S GPUs. For LEAP<sub>OP1</sub>, the historical sequence lengths evaluated are  $l_1 \in \{3, 7\}$  days. For LEAP<sub>OP2</sub>, the few-shot prompt uses  $l_2 = 5$  quintuples as in-context examples. For LEAP<sub>MEF</sub>, the default historical window is set to  $l_3 = 7$  days. Detailed hyper-parameter configurations and fine-tuning procedures for the RoBERTa<sub>BASE</sub> encoder and FLAN-T5<sub>BASE</sub> generator are provided in the published GitHub repository, where LEAP’s source code and sampled data for OP and MEF training and evaluation are also included to ensure reproducibility. An empirical computational cost analysis, including training time, memory occupation, and estimated commercialized LLM API budget is delivered in supplementary material Appendix D.

## 4.2 LEAP<sub>OP1</sub>: Ranking Results

We evaluate ranking-based OP using Hits@ $k$  ( $k \in \{1, 3, 10\}$ ), which measures the proportion of test queries for which the correct object appears within the top- $k$  ranked candidates. Baselines include decoder-only ConvTransE [33], GNN-based CompGCN [38], R-GCN [32], and RE-GCN [18], as well as hypergraph-based SeCoGD [23]. For fair comparison, all structural methods employ ConvTransE as the scoring decoder. Table 2 presents the ranking OP results, where LEAP<sub>OP1</sub>

**Table 2.** Ranking OP results. **Bold:** Best. Underline: Second best per column.

| Method                                    | Afghanistan |             |             | India       |             |             | Russia      |             |             |
|-------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                           | Hits@1      | H@3         | H@10        | H@1         | H@3         | H@10        | H@1         | H@3         | H@10        |
| Historical sequence length $l_1 = 3$ days |             |             |             |             |             |             |             |             |             |
| ConvTransE                                | .124        | .270        | .492        | .152        | .282        | .460        | .101        | .179        | .308        |
| CompGCN                                   | .121        | .263        | .490        | .170        | .314        | .494        | .117        | .189        | .312        |
| R-GCN                                     | .144        | .313        | .554        | .172        | .321        | .501        | .131        | .239        | .390        |
| RE-GCN                                    | .154        | .314        | .541        | .170        | .313        | .495        | .133        | .235        | .377        |
| SeCoGD                                    | <u>.188</u> | <u>.357</u> | <u>.574</u> | <u>.206</u> | <u>.355</u> | <u>.536</u> | <u>.177</u> | <u>.291</u> | <u>.435</u> |
| LEAP <sub>OP1</sub>                       | <b>.369</b> | <b>.563</b> | <b>.732</b> | <b>.368</b> | <b>.551</b> | <b>.723</b> | <b>.375</b> | <b>.539</b> | <b>.683</b> |
| Historical sequence length $l_1 = 7$ days |             |             |             |             |             |             |             |             |             |
| ConvTransE                                | .124        | .276        | .491        | .154        | .291        | .474        | .097        | .180        | .297        |
| CompGCN                                   | .109        | .273        | .508        | .166        | .312        | .499        | .107        | .185        | .316        |
| R-GCN                                     | .158        | .324        | .553        | .176        | .321        | .500        | .128        | .246        | .393        |
| RE-GCN                                    | .155        | .330        | .554        | .172        | .318        | .500        | .134        | .235        | .378        |
| SeCoGD                                    | <u>.183</u> | <u>.365</u> | <u>.586</u> | <u>.206</u> | <u>.352</u> | <u>.535</u> | <u>.166</u> | <u>.282</u> | <u>.443</u> |
| LEAP <sub>OP1</sub>                       | <b>.386</b> | <b>.588</b> | <b>.766</b> | <b>.394</b> | <b>.583</b> | <b>.745</b> | <b>.386</b> | <b>.559</b> | <b>.708</b> |

achieves substantial improvements over all baselines across the three datasets and both sequence lengths. At  $l_1 = 7$ , LEAP<sub>OP1</sub> improves Hits@1 by absolute margins of 20.3%, 18.8%, and 22.0% over the strongest baseline (SeCoGD) on Afghanistan, India, and Russia, respectively. These gains are consistent across all Hits@ $k$  levels, confirming that the integration of domain-adapted LLM-based semantic embeddings provides a stronger contextual signal than typical topic clustering identifiers [5]. Additionally, extending the historical window from  $l_1 = 3$  to  $l_1 = 7$  yields modest but consistent Hits score improvements, suggesting that longer historical context is beneficial to ranking OP.

### 4.3 LEAP<sub>OP2</sub>: Generative Results

We evaluate generative OP using ROUGE scores [20], calculating ROUGE-1 (R-1), ROUGE-2 (R-2), and ROUGE-L (R-L) between the generated text and the ground-truth object entity name. We adopt ROUGE rather than exact matching because socio-political entity names often admit surface-level variation, including abbreviations and alternative transliterations. We acknowledge that ROUGE cannot resolve deeper semantic equivalences, such as “Government (AFG)” vs. “Government of Afghanistan”; incorporating entity-level normalization or embedding-based similarity metrics is a worthwhile direction for future work. We compare against selected competitive structural baselines, SeCoGD and LEAP<sub>OP1</sub>, by reading out the highest-probability entity name as the generated answer. For LLM-only baselines, we directly prompt selected LLMs without task-specific fine-tuning. The direct-prompting baselines span two categories: (i) open-source models including Mistral-7B-Instruct [15], Llama-3.1-8B-

**Table 3.** Generative OP results. **Bold:** Best. Underline: Second best per column. Closed-source LLMs and DeepSeek-V3.2 are evaluated on only the first 5,000 test samples due to cost constraints in purchasing API tokens rather than local deployment.

| Method                                                                          | Afghanistan |             |             | India       |             |             | Russia      |             |             |
|---------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                                                                 | R-1         | R-2         | R-L         | R-1         | R-2         | R-L         | R-1         | R-2         | R-L         |
| Structural readout, fine-tuned RoBERTa <sub>BASE</sub> encoder, $l_1 = 3$ days  |             |             |             |             |             |             |             |             |             |
| SeCoGD                                                                          | .427        | .166        | .427        | .481        | .290        | .481        | .370        | .141        | .370        |
| LEAP <sub>OP1</sub>                                                             | .529        | .267        | .529        | .550        | .382        | .550        | .497        | .263        | .497        |
| Structural readout, fine-tuned RoBERTa <sub>BASE</sub> encoder, $l_1 = 7$ days  |             |             |             |             |             |             |             |             |             |
| SeCoGD                                                                          | .430        | .172        | .430        | .483        | .291        | .483        | .383        | .140        | .383        |
| LEAP <sub>OP1</sub>                                                             | .568        | .320        | .567        | .589        | .403        | .589        | .524        | .264        | .524        |
| Open-source LLMs, few-shot prompt, $l_2 = 5$ quintuples                         |             |             |             |             |             |             |             |             |             |
| Mistral-7B                                                                      | .366        | .110        | .362        | .342        | .132        | .342        | .346        | .118        | .345        |
| Llama-3.1-8B                                                                    | .311        | .109        | .310        | .280        | .127        | .280        | .393        | .158        | .393        |
| DeepSeek-V3.2                                                                   | .510        | .197        | .509        | .437        | .244        | .436        | .479        | .217        | .479        |
| Closed-source LLMs, few-shot prompt, $l_2 = 5$ quintuples                       |             |             |             |             |             |             |             |             |             |
| GPT-3.5 Turbo                                                                   | .410        | .130        | .409        | .364        | .160        | .364        | .348        | .126        | .349        |
| GPT-4o mini                                                                     | .415        | .130        | .413        | .384        | .177        | .383        | .388        | .173        | .388        |
| GPT-4o                                                                          | .493        | .193        | .491        | .417        | .213        | .417        | .466        | .213        | .466        |
| GPT-5.2                                                                         | .518        | .220        | .516        | .458        | .270        | .458        | .496        | .240        | .496        |
| LEAP <sub>OP2</sub> : Fine-tuned FLAN-T5 <sub>BASE</sub> , $l_2 = 5$ quintuples |             |             |             |             |             |             |             |             |             |
| No-text                                                                         | .422        | .132        | .422        | .478        | .252        | .478        | .361        | .101        | .361        |
| Zero-shot                                                                       | <u>.860</u> | <u>.560</u> | <u>.860</u> | <u>.853</u> | <u>.689</u> | <u>.853</u> | <u>.832</u> | <u>.448</u> | <u>.832</u> |
| Few-shot                                                                        | <b>.864</b> | <b>.566</b> | <b>.864</b> | <b>.859</b> | <b>.696</b> | <b>.859</b> | <b>.842</b> | <b>.454</b> | <b>.841</b> |

Instruct [12], and DeepSeek-V3.2 [21]; (ii) closed-source models including GPT-3.5-Turbo-Instruct [28], GPT-4o [25], GPT-4o mini [24], and GPT-5.2 [26].

Based on Table 3, we discuss four notable observations on the generative OP results. First, LEAP<sub>OP2</sub> with the few-shot prompt achieves the highest ROUGE scores across all datasets, with R-1 exceeding 0.80 on all three subsets. Second, the zero-shot variant of LEAP<sub>OP2</sub> performs comparably to the few-shot variant, demonstrating that instruction fine-tuning builds up robust in-domain generalization in FLAN-T5<sub>BASE</sub>, even without in-context examples. Third, the no-text ablation exhibits significant degradation across all datasets, confirming that textual summaries provide essential semantic context for accurate question answering. Fourth, all direct-prompting LLMs, including the most recent GPT-5.2, explicitly underperformed the fine-tuned LEAP<sub>OP2</sub>. A clear scaling trend where GPT-5.2 consistently outperforms GPT-4o, which in turn outperforms GPT-4o mini, is also noticeable. Similarly, DeepSeek-V3.2 surpasses other open-source LLMs. These observations highlight a practical insight: under comparable few-shot prompting conditions, fine-tuning a compact model on task-specific data outperforms prompting larger general-purpose models. We note that this comparison is inherently asymmetric: the fine-tuned models benefit

**Table 4.** MEF results ( $l_3 = 7$  days). **Bold:** Best. Underline: Second best per column. RoB.: RoBERTa<sub>LARGE</sub>. DeB.: DeBERTaV3<sub>LARGE</sub>. SA: Self-attention. Ablation studies on LEAP<sub>MEF</sub>’s temporal aggregation mechanisms are also included by replacing SA and LSTM layers with simple mean and max pooling.

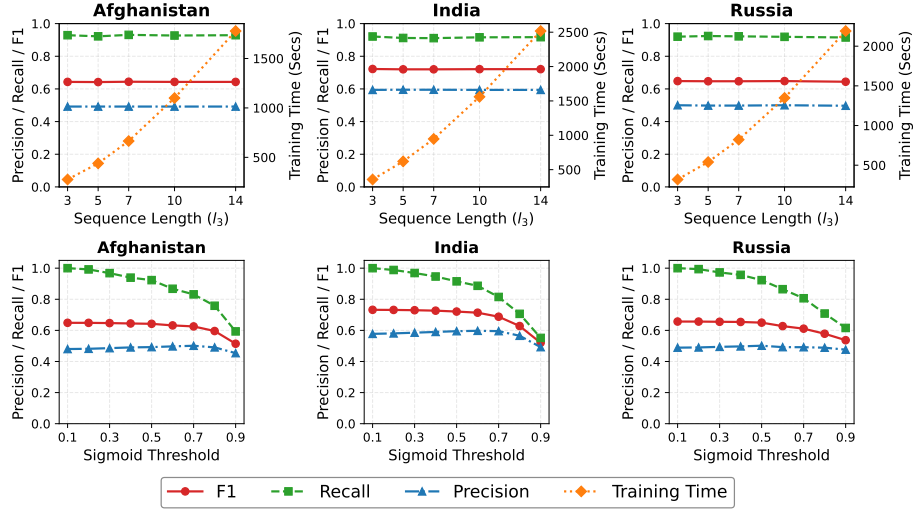
| Method            | Afghanistan |             |             | India       |             |             | Russia      |             |             |
|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                   | F1          | R           | P           | F1          | R           | P           | F1          | R           | P           |
| MLARAM            | .338        | .353        | .324        | .337        | .341        | .333        | .257        | .267        | .248        |
| BR-KNN            | .499        | .615        | .420        | .504        | .560        | .458        | .475        | .566        | .409        |
| ML-KNN            | .454        | .504        | .413        | .523        | .558        | .492        | .514        | .586        | .458        |
| DNN               | .558        | .681        | .473        | .525        | .564        | .491        | .538        | .626        | .472        |
| DynamicGCN        | .501        | .578        | .442        | .418        | .432        | .405        | .528        | .601        | .471        |
| TemporalGCN       | .600        | .769        | .492        | .607        | .672        | .553        | .564        | .677        | .483        |
| RENET             | .606        | .778        | <u>.496</u> | .584        | .642        | .536        | .559        | .657        | .486        |
| Glean             | .625        | .828        | <b>.502</b> | .667        | .773        | .587        | .589        | .736        | .491        |
| LEAP (RoB., Mean) | .611        | .792        | <u>.497</u> | .593        | .652        | .544        | .566        | .687        | .481        |
| LEAP (DeB., Mean) | .611        | .789        | <b>.499</b> | .598        | .656        | .549        | .565        | .679        | .484        |
| LEAP (RoB., Max)  | .609        | .782        | <b>.499</b> | .598        | .656        | .549        | .568        | .688        | .484        |
| LEAP (DeB., Max)  | .607        | .790        | .493        | .613        | .681        | .557        | .563        | .676        | .482        |
| LEAP (RoB., SA)   | .636        | .887        | <u>.496</u> | <u>.710</u> | .873        | <b>.598</b> | .628        | .868        | .492        |
| LEAP (DeB., SA)   | <u>.637</u> | .892        | .495        | <u>.710</u> | .874        | <b>.598</b> | .628        | .868        | .492        |
| LEAP (RoB., LSTM) | <b>.644</b> | <b>.934</b> | .491        | <b>.720</b> | <b>.912</b> | <u>.595</u> | <u>.646</u> | <u>.921</u> | <u>.497</u> |
| LEAP (DeB., LSTM) | <b>.644</b> | <u>.931</u> | .492        | <b>.720</b> | <u>.911</u> | <u>.595</u> | <b>.647</b> | <b>.922</b> | <b>.498</b> |

from full training-set exposure whereas the prompted models see only five examples, and the gap may narrow with more sophisticated prompting strategies such as chain-of-thought reasoning or retrieval-augmented generation. Nevertheless, the result underscores that for narrow, well-defined prediction tasks with ample training data, task-specific adaptation remains a cost-effective alternative to frontier model scaling. The generative formulation is suitable for OP as socio-political entities tend to be short phrases, keeping the output space manageable for a relatively small encoder-decoder language model like FLAN-T5<sub>BASE</sub>.

#### 4.4 LEAP<sub>MEF</sub>: Multi-Event Forecasting Results

We evaluate MEF using precision (P), recall (R), and F1 score for the multi-label binary classification of possible relation occurrences into next-day extrapolation. Baselines include neural network-based MLARAM [4] and DNN [3], nearest-neighbor methods BR-KNN [37] and ML-KNN [43], GNN/RNN-based DynamicGCN [10], TemporalGCN [45], and RENET [16], as well as the text-aware Glean [11]. We report four LEAP<sub>MEF</sub> configurations that combine two quintuple-level prompt encoders (RoBERTa<sub>LARGE</sub> and DeBERTaV3<sub>LARGE</sub>) with two key temporal aggregators (self-attention and LSTM). The mean and max pooling options are ablation studies on temporal aggregation mechanism.

Based on Table 4, we discuss three salient observations. First,  $\text{LEAP}_{\text{MEF}}$  with LSTM aggregation achieves the highest F1 score across all three datasets, with significant recall improvements over Glean. These substantial recall gains indicate that  $\text{LEAP}_{\text{MEF}}$  is effective at identifying relations that will occur, which is the more practically valuable capability in early-warning scenarios. Second, the eight  $\text{LEAP}_{\text{MEF}}$  configurations in Table 4 reveal a clear aggregator ranking: LSTM consistently achieves the highest F1 and recall, followed by self-attention, while mean and max pooling as ablation studies yield notably lower performance. The significant recall gap between LSTM and mean/max pooling also demonstrates that position-invariant pooling discards critical sequential information and that order-aware aggregation is essential for capturing temporal dynamics in event histories. Third, the two encoders produce closely matched results under the same aggregation strategy, demonstrating the encoder-agnostic nature of the architecture. DeBERTaV3<sub>LARGE</sub> with LSTM achieves marginally higher F1 on Russia and ties on Afghanistan and India, suggesting that the improved pre-training objectives of DeBERTaV3 provide slightly richer quintuple representations than RoBERTa’s masked language modeling alone. All  $\text{LEAP}_{\text{MEF}}$  variants employ neither GNNs nor RNNs for graph-level encoding, which substantially reduces architectural complexity compared to hybrid baselines.



**Fig. 2.** Sensitivity analyses on  $\text{LEAP}_{\text{MEF}}$  (DeBERTaV3<sub>LARGE</sub>, LSTM). Top row: Effect of sequence length  $l_3$  (defaulting to 7) on precision, recall, and F1 (left y-axis) and one-epoch training time in seconds (right y-axis). Bottom row: Effect of element-wise Sigmoid decision threshold (defaulting to 0.5) on precision, recall, and F1.

#### 4.5 LEAP<sub>MEF</sub>: Sensitivity Analyses

*Sensitivity Analysis on Sequence Length.* To assess the robustness of LEAP<sub>MEF</sub> (DeBERTaV3<sub>LARGE</sub>, LSTM) with respect to the historical window length, we vary  $l_3 \in \{3, 5, 7, 10, 14\}$  and report the results in Figure 2. Across all three ICEWS subsets, precision, recall, and F1 scores remain remarkably stable as the sequence length increases. For Afghanistan, F1 varies within a narrow range of only 0.002 (from .642 to .644), and similarly tight ranges are observed for India and Russia. Meanwhile, training time grows approximately linearly with  $l_3$ , rising from roughly 5–6 minutes per epoch at  $l_3 = 3$  to 30–45 minutes at  $l_3 = 14$  depending on the dataset. This stability suggests that the LSTM aggregation effectively captures the relevant temporal patterns from even short histories.

Interestingly, this behavior contrasts with the interpolated OP task. As shown in Tables 2 and 3, extending the historical window from  $l_1 = 3$  to  $l_1 = 7$  in LEAP<sub>OP1</sub> yields consistent Hits@ $k$  improvements across all datasets, for instance, Hits@1 increases by 1.7%, 2.6%, and 1.1% absolute on Afghanistan, India, and Russia, respectively, with corresponding gains in the structural readout ROUGE scores of LEAP<sub>OP2</sub>. We attribute this divergence to the fundamentally different characteristics of the two tasks. Object prediction requires resolving a specific missing entity for a given query, where richer historical context provides more evidence about entity interaction patterns and co-occurrence statistics, directly benefiting the ranking or generation of the correct object. In contrast, multi-event forecasting predicts which relation could potentially occur at a future timestamp, which is a coarser-grained classification task over the relation vocabulary. The distribution of active relations tends to exhibit greater temporal regularity than individual entity-level interactions, meaning that a few recent days already capture the prevailing relational patterns. Since both tasks incur increased computational cost with longer sequences, the practical implication for MEF is that researchers can adopt a shorter window, such as  $l_3 = 3$ , to reduce training time without sacrificing forecasting quality, whereas for OP, investing in longer histories is more justified by the corresponding accuracy gains.

*Sensitivity Analysis on Decision Threshold.* We further examine the sensitivity of LEAP<sub>MEF</sub> (DeBERTaV3<sub>LARGE</sub>, LSTM,  $l_3 = 7$ ) by varying the element-wise Sigmoid threshold, used to classify relation occurrences, across  $\{0.1, 0.2, \dots, 0.9\}$ . Figure 2 also reports the results on all three ICEWS subsets. As the threshold increases, recall decreases monotonically since fewer relations pass the stricter decision boundary, while precision increases modestly at first, on an approximately absolute scale of 0.02, as false positives are filtered out, then declines at very high thresholds where the model begins to miss even confidently predicted relations. The F1 score remains stable across the low-to-moderate threshold range before gradually deteriorating at higher values, indicating that the model’s predicted probabilities are well calibrated and that the ranking of relation scores is robust to the exact choice of binarization point. The continuously decreasing recall given higher thresholds also indicates that the majority of relations are negative on any given day, so a more conservative threshold naturally reduces

**Table 5.** Error analyses on LEAP<sub>MEF</sub> (RoBERTa<sub>LARGE</sub>, LSTM,  $l_3 = 7$ ).  $\bar{n}_{\text{pred}} / \bar{n}_{\text{true}}$ : mean number of daily predicted / truly active relations. FP / (FP + FN): proportion of all errors that are false positives.  $\rho_{\text{freq}}$ : Spearman correlation [36] between a relation’s training frequency (freq) and its per-relation F1.  $\rho_{\text{day}}$ : Spearman correlation between daily count of active relations and daily F1. Q1 / Q4 F1: mean per-relation test F1 for the lowest / highest training-frequency quartile (Q) among test-active relations.

|                                                 | Afghanistan | India   | Russia  |
|-------------------------------------------------|-------------|---------|---------|
| $\bar{n}_{\text{pred}} / \bar{n}_{\text{true}}$ | 83 / 29     | 88 / 49 | 89 / 32 |
| min / max $n_{\text{true}}$                     | 9 / 51      | 25 / 68 | 8 / 58  |
| FP / (FP + FN)                                  | .967        | .908    | .960    |
| $\rho_{\text{freq}}$                            | .918        | .899    | .896    |
| $\rho_{\text{day}}$                             | .908        | .830    | .957    |
| Q1 (low-frequency) F1                           | .000        | .000    | .000    |
| Q4 (high-frequency) F1                          | .657        | .817    | .616    |

the overall number of misclassified labels. The default threshold of 0.5 strikes an effective balance, achieving near-peak F1 while maintaining high recall, which is the more operationally critical metric in early-warning applications where failing to anticipate an emerging relation type carries greater cost than issuing a false alarm. We observe consistent patterns across all three country subsets, suggesting that these threshold dynamics are not dataset-specific but rather reflect a general property of the multi-label forecasting formulation.

#### 4.6 LEAP<sub>MEF</sub>: Error Analyses

To complement the metrics in Table 4, we conduct detailed error analyses for LEAP<sub>MEF</sub>, using RoBERTa<sub>LARGE</sub> encoder, LSTM aggregator, and historical sequence length  $l_3 = 7$ , on all three ICEWS test sets. For each test day, we collect all relations’ element-wise Sigmoid-activated probabilities and the ground-truth binary labels. We then compute relation-specific diagnostics across all test days. Table 5 summarizes the key findings of error analyses on LEAP<sub>MEF</sub>.

The dominant error mode across all three subsets is over-prediction. False positives account for over 90% of all classification errors (96.7% on AFG, 90.8% on IND, and 96.0% on RUS), while false negatives constitute less than 10%. This asymmetry is consistent with the high recall but comparatively lower precision reported in Table 4. A closer examination reveals the underlying mechanism: the model predicts a near-constant number of relations per test day (83 on AFG, 88 on IND, and 89 on RUS), whereas the number of truly active relations varies substantially across days, such as 9 to 51 on AFG, 25 to 68 on IND, and 8 to 58 on RUS. LEAP<sub>MEF</sub> does effectively learn a stable “default active set” of relations from training-time patterns but does not adapt its prediction count to individual days. On days with fewer active relations, the gap between the predicted and the true sets widens, producing a large number of false positives and driving down the daily F1 score. This gap is also demonstrated by the strong

Spearman correlation [36] between the daily count of active relations and the daily F1, where  $\rho_{\text{day}}$  ranges from .830 to .957 across all three subsets, indicating that prediction quality is tightly coupled with daily relation density.

Another salient finding is a strong monotonic relationship between a relation’s training frequency and its test-time prediction quality, where the Spearman correlations  $\rho_{\text{freq}}$  [36] exceeds .890 on all three subsets. When test-active relations are grouped into four training-frequency quartiles, the lowest quartile (Q1) achieves a mean F1 of exactly .000 across all subsets, meaning that no low-frequency relation is ever correctly predicted. In contrast, the highest quartile (Q4) reaches mean F1 scores of .657 (AFG), .817 (IND), and .616 (RUS). In specific, high-frequency relations such as “Make statement” and “Consult” are predicted with near-perfect F1 ( $\sim .99$ ), while mid-frequency relations with 50 to 150 training occurrences such as “Apologize” and “Express intent to settle dispute” yield zero true positives despite appearing on 5 to 15 test days each. This pattern suggests that the LSTM aggregator effectively captures temporal regularities of frequently recurring relations but receives insufficient gradient signal to learn discriminative representations for those long-tail relations.

## 5 Conclusion

We propose LEAP, a systematic framework that leverages large language models for both interpolated and extrapolated event predictions on temporal knowledge graphs. Through three complementary components, we demonstrate that compact open-source LLMs, when equipped with task-specific instruction fine-tuning and carefully designed prompt templates, can match or exceed the performance of closed-source commercialized LLMs and outperform conventional GNN/RNN-based frameworks. The encoder/aggregator-agnostic design of LEAP<sub>MEF</sub> further establishes LEAP’s modularity, with ablation studies revealing that LSTM as the temporal aggregator combined with DeBERTaV3<sub>LARGE</sub> as the quintuple-level prompt encoder yields the strongest forecasting performance. Future work could extend LEAP to subject/relation/timestamp prediction, multi-step forecasting jumping out of the next-day horizon, and TKG datasets from domains beyond socio-political events to explore its broader applicability.

## Use of Generative AI

The authors used Claude Opus 4.6 [1] to assist with correcting grammar and refining readability. The authors take full responsibility for the final manuscript.

## Ethics Considerations

The ICEWS datasets used in this work contain structured and semantic records of interactions between socio-political entities and organizations. Our framework, LEAP, is intended to support academic research and early-warning applications, and we encourage responsible deployment with appropriate human oversight.

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